

Aniruddh Chandratre

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0-1A (current) · EB-1A approved

ML infrastructure engineer specializing in distributed data platforms, GPU scheduling, and cluster efficiency. Builds production systems for FSD, Optimus, and Digital Optimus across large heterogeneous GPU fleets.

EXPERIENCE

Senior Engineer, ML Infrastructure — Tesla AI Feb 2024 – Present · Palo Alto, CA
[Distributed data platform](#) · [GPU scheduling](#) · [cluster efficiency](#)

- Lead Tesla AI's internal data processing platform for acquisition, curation, mining, auto-labeling, training, and evaluation; it runs **10B+ jobs** and **3,000+ compute-years/month** across **10** GPU clusters.
- Re-architected the platform core, distributed worker coordination, and memory model, increasing throughput from **940M to 15B+** data items/month and maximum dataset size from **1M to 1B+** clips per run.
- Built scheduling and live defragmentation for a heterogeneous GPU fleet, increasing usable compute capacity from **24% to 88%** and fleet-wide effective utilization to **98%**.
- Added end-to-end lineage across artifacts, datasets, training jobs, and models for multi-stage pipeline provenance.

Software Engineer, Automation Software — Tesla Jan 2023 – Jan 2024 · Fremont, CA
[Edge-to-cloud data infrastructure for the gigafactories](#)

- Designed and shipped an edge-to-cloud data broker processing **10B+ manufacturing data points/day** across Austin, Berlin, and Fremont.
- Developed an mmap-backed durable queue sustaining **5M writes/s** on NFS-backed Kubernetes volumes and a distributed cache processing **10M entries/s**, cutting storage **80%** across **50** deployments.
- Built and operated **20** high-availability services for factory insights.

Graduate Research Assistant — ASU, Cyber-Physical Systems Lab Jan 2021 – Dec 2022 · Tempe, AZ

- Researched formal verification under DARPA ARCOS with Lockheed Martin; co-authored PSY-TaLiRo and operated the Kubernetes cluster for high-volume simulation workloads.

Founding ML Engineer — AfterShoot Aug 2020 – Jan 2021 · Remote

- Built the image-ranking model at the core of AfterShoot's photo-culling product, now used by approximately **250K** photographers.

EDUCATION

M.S. Robotics & Autonomous Systems (AI) — Arizona State University · 4.0/4.0, Distinction Medal 2022

B.Tech Electronics & Telecommunications — NMIMS MPSTME, Mumbai · Dean's List 2020

RESEARCH, AWARDS & TOOLING

PUBLICATIONS 8 peer-reviewed papers · 174 citations · h-index 5 — HSCC, EMSOFT, FMICS, ARCH. First author, HSCC 2023: *Stealthy Attacks Formalized as STL Formulas for Falsification of CPS Security*.

AWARDS 5× national hackathon winner in India — Smart India Hackathon 2019, Mastek Deep Blue (×2), IET Hack N Code, RAENG.

LANGUAGES Go, Python, C, C++, JavaScript/TypeScript, Java

SYSTEMS Slurm, Kubernetes, Helm, Docker, AWS, GCP, Prometheus, Grafana, Kafka, ClickHouse, InfluxDB, MQTT, SQL